

Strength-Parity Ensembling with Parameter-Isolated Experts for Multi-Task Affect Recognition

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Abstract—Leading entries on the multi-task track of the 11th ABAW challenge rely on heavy ensembling, yet which member is worth adding to an already strong ensemble is rarely made explicit. We study this question for joint valence–arousal estimation, 8-way expression recognition, and 12-way action-unit detection from a single unconstrained face, under partial, long-tailed labels and a rule that forbids pretraining on Aff-Wild2. Building on a shared affect-latent that marginalizes the missing labels across two affect-supervised backbones, we propose a strength-parity rule: an added member lowers the ensemble error only when it is both decorrelated from the current members and a near-peer of them in individual accuracy. The rule exposes a concrete obstacle, as on a single backbone re-seeding and even distinct fine-tuning curricula re-converge to a prediction correlation of 0.98 and add no diversity. Parameter-isolation removes it: confining each adaptation to a disjoint low-rank subspace of a shared backbone yields experts that stay decorrelated at 0.91 while remaining near-peers, the strongest of them an AffectNet-adapted expert. The resulting system raises the overall validation score to 1.6949, against the organizers’ ConvNeXt-with-MixAugment baseline of 0.45; with per-AU calibration and by pooling the shared-latent heads’ valence–arousal byproduct as a further near-peer, the best configuration reaches 1.7259.

Index Terms—Affective computing, multi-task learning, model ensembling, parameter-efficient fine-tuning, ABAW.

I. INTRODUCTION

Recognizing emotion from an in-the-wild face is not a single problem but three that compete with one another. The 11th Affective Behavior Analysis in-the-wild (ABAW) challenge [2], [3] makes this tension explicit in its multi-task learning (MTL) track: from a single cropped face on the s-Aff-Wild2 corpus, a model must place the affect on the continuous valence–arousal circumplex, name one of eight categorical expressions, and flag which of twelve facial action units are active. The three label families are annotated on overlapping but unequal frame subsets, are individually long-tailed, and the challenge forbids any use of Aff-Wild2 for pretraining. The ABAW series [2]–[7] has driven progress on this corpus, and leading MTL entries share a recipe: a face-pretrained transformer backbone, task-specific heads, and heavy ensembling [8]–[10], with the current published validation figure set by the Progressive Learning system [11]. The recurring difficulty is not the individual tasks but their partial and imbalanced supervision; the standard responses mask the missing labels in the loss or impute them with a teacher [12], [13].

A base system for this track already exists, and it takes a different route to the missing labels. Companion work [1]

casts the partial-label problem as marginalization over a shared affect-latent (SAL): a small variational bottleneck mediates the three task decoders, so that a frame carrying only one label still shapes the representation the other two rely on [14], and pairs the latent with two affect-supervised face backbones whose per-task predictions are averaged. We adopt that system as our starting point rather than revisiting it.

Ensembling and its dependence on member diversity are long understood [15]–[17], but the ABAW literature typically diversifies by backbone or by seed and reports the pooled number. Our question is narrower and operational: given this fixed strong ensemble, we ask when adding one more member helps, and what property that member must possess.

The answer, our core contribution, is a compact admission criterion. An added member helps only when it satisfies two conditions simultaneously: it must disagree with the current ensemble (be decorrelated) and be about as accurate as the members already in it (a near-peer). We call this criterion strength-parity. Stated in the abstract, it is nearly self-evident; what is not, and what matters in practice, is which of the inexpensive ways of producing a “new” model in fact satisfy it. Re-seeding, and even running a different supervised curriculum on the same backbone, yield nearly the same predictions; distinct full fine-tuning curricula on one backbone correlate at 0.98, and therefore add nothing to average over. Decorrelation is a property of the pretrained feature space rather than of the path taken to fine-tune it.

This diagnosis is also constructive: if unconstrained fine-tuning re-converges, constraining it should not. We adapt a single shared backbone to different affect corpora through disjoint low-rank subspaces, in the manner of LoRA [18] but repurposed from a parameter-efficient adapter into a diversity mechanism [19], and the resulting experts stay decorrelated, at a correlation of 0.91 among experts on one backbone, while each remains a near-peer. An AffectNet-adapted low-rank expert is the strongest single expression member in the pool, at macro-F1 0.4762, above either base backbone, and adding it raises the ensemble expression score from 0.4698 to 0.4978 and the overall validation score to 1.6949. This supplies the mechanism the strength-parity rule requires: a means of producing decorrelated near-peers from a single backbone, without a second pretrained one. In summary, our contributions are as follows

- A strength-parity rule for ensembling partially-labeled affect models: a member helps only if it is both decorrelated

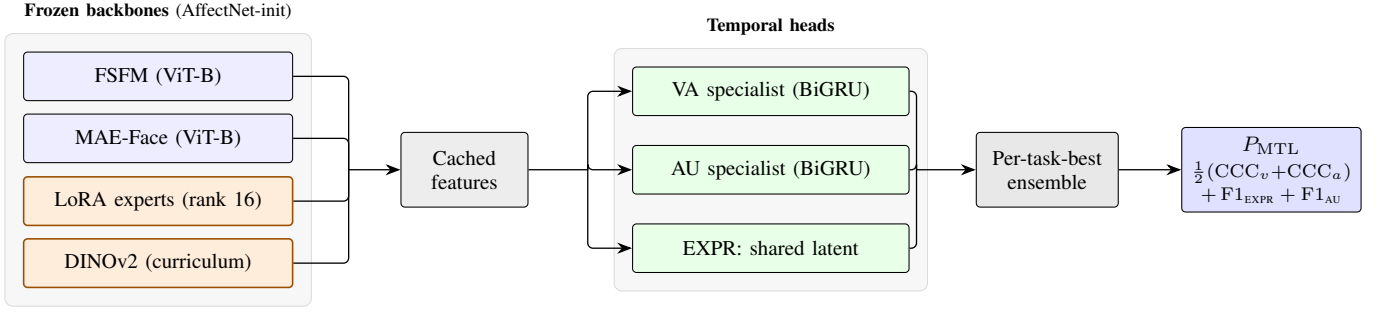


Fig. 1. System overview. Frozen affect-supervised backbones and the added members (highlighted: parameter-isolated LoRA experts and a curriculum near-peer) supply cached per-frame features to BiGRU specialists (valence–arousal, action units) and a shared affect-latent (expression), whose per-task-best members are averaged. The shared latent and two-backbone base follow companion work [1].

from the ensemble and an individual near-peer, which we use as a pre-ensemble admission test.

- A mechanism that supplies such members: low-rank adaptation of a single backbone yields decorrelated near-peer experts where full fine-tuning re-converges, which explains why cheaper diversity sources fail.
- Results on the 11th ABAW MTL track: the system reaches 1.6949 on validation and 1.7259 in its strongest configuration, against a baseline of 0.45, with every gain sized by a paired video-level bootstrap.

II. METHOD

Fig. 1 gives an overview of the full pipeline; we first summarize the base recognition system we build on, then state the rule that governs which additional members improve it and the mechanism that produces such members. The base system’s design, ablations, and justification belong to the companion work [1] and are not claimed here.

A. The shared affect-latent base system

Backbones and features. Two ViT-B/16 face backbones supply complementary representations: one is self-supervised on VGGFace2 in the FSFM manner [20], the other masked-autoencoded on a large face corpus in the MAE-Face manner [21]. Each passes through the same three stages: categorical fine-tuning on AffectNet [22] to make the features affect-aware, a short per-frame adaptation on s-Aff-Wild2 with the top blocks unfrozen, and a freeze, after which the network is a fixed extractor whose per-frame 768-dimensional embeddings are cached once. Every temporal model then reads these cached features instead of back-propagating into the backbone, which is what makes an individual member cheap enough to train and to ensemble at the scale the rest of the paper relies on.

Temporal heads and the shared latent. Over a window of cached features, a bidirectional GRU produces per-frame states that feed three task decoders. Valence and arousal are regressed by a dedicated head under a concordance loss, and the twelve action units by a multi-label head under binary cross-entropy; each loss is evaluated only on the frames that carry the corresponding label, so a partially annotated frame still trains

the tasks it can. Expression is decoded through the shared affect-latent SAL: a variational bottleneck mediates all three decoders and is trained by a masked evidence lower bound [1], [14] that marginalizes each frame’s absent labels, so that a frame carrying only action-unit labels still updates the latent the expression decoder reads. Routing expression through this shared latent, rather than a task-specific head, yields per-backbone macro-F1 of 0.4416–0.4461 and outperforms a matched masked-loss joint model.

The two backbones are individually near-peers and make different errors, so averaging their per-task predictions improves every task; assembled per-task-best, the base system reaches 1.6669 on validation. Its expression score, however, rests on only two shared-latent members, and an average is no better than its members are diverse. The remainder of this section adds a member that improves it.

B. The strength-parity rule

Let the current ensemble average the per-frame posteriors of its M members,

$$\bar{p} = \frac{1}{M} \sum_{i=1}^M p_{m_i}, \quad (1)$$

and consider adding one more. The averaged error of such an ensemble follows the classic accuracy–diversity decomposition [15],

$$\mathbb{E} \|\bar{e}\|^2 = \varepsilon^2 \left(\bar{\rho} + \frac{1 - \bar{\rho}}{M} \right), \quad (2)$$

where ε^2 is the mean squared error of the members and $\bar{\rho}$ their mean pairwise correlation. The average therefore improves only when the new member lowers the correlation without raising the error level: a candidate that duplicates the ensemble leaves the correlation untouched and contributes nothing, however accurate, whereas a decorrelated but weak candidate raises the error level faster than it lowers the correlation. The useful region is the intersection of the two, decorrelated and near-peer, and we treat membership in it, judged from the correlation of a candidate’s posteriors with the ensemble and from its own task metric, as a pre-ensemble admission test rather than a property discovered after pooling.

The two conditions are genuinely separable: a re-seeded head on one backbone is strong but redundant, while a general-purpose backbone is decorrelated but too weak for affect. A useful member must satisfy both conditions, and the rest of this section builds such members, first through a supervised curriculum and then, more cheaply, through parameter-isolation.

C. A curriculum-built near-peer

The natural source of decorrelation is a third backbone with a different pretraining, and the obvious candidate is DINOv2 [23], whose general self-supervised features are the most decorrelated from our face backbones. General pretraining alone is too weak for affect, so we give it a supervised curriculum. Adapting DINOv2 through AffectNet and then EmotionNet [24] action-unit supervision before the s-Aff-Wild2 stage turns it into an overall per-frame near-peer of the face backbones, and its shared-latent expression member reaches macro-F1 0.4540 while staying decorrelated from them, a member that satisfies both conditions. The curriculum is chosen to match the in-the-wild target distribution rather than to maximize accuracy on any single source corpus.

D. Parameter-isolated experts

The curriculum gives one near-peer, but each new one costs a full pretraining chain, and the deeper problem remains: several full fine-tuning curricula on the same backbone re-converge to nearly the same predictions, correlated at 0.98, so they cannot be pooled with one another. Unconstrained fine-tuning erases the diversity we set out to create.

Constraining the adaptation preserves it: we freeze a shared near-peer backbone and adapt it to different affect corpora through disjoint low-rank residual subspaces [18], one expert per corpus. Each expert replaces a frozen projection weight W_0 by

$$W = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times k}, \quad (3)$$

with rank 16 and scaling factor 32, on the attention and MLP projections; the shared weight stays frozen, and only the low-rank factors differ between experts. Because the experts occupy different low-rank subspaces of the same frozen features, they specialize without collapsing onto one another, and two comparisons make this concrete. On a single backbone, isolated experts sit at a prediction correlation of 0.91, below the 0.98 to which full fine-tuning re-converges, so the constraint prevents same-backbone adaptations from collapsing to a single model. More importantly for our purpose, the strongest expert is genuinely decorrelated from the two base backbones it joins, at 0.75, comparable to the 0.81 between the two backbones themselves. Strength is preserved as well: the AffectNet-adapted expert reaches expression macro-F1 0.4762, above both base backbones and above the curriculum member, and is the strongest single expression member in the pool. Parameter-isolation thus supplies what the strength-parity rule requires, decorrelated near-peers produced from a single backbone, where full fine-tuning could yield only decorrelated-or-strong, never both.

E. Assembly and calibration

The final prediction is assembled per task from the members that are strongest on that task, all averaged at the posterior level. Valence–arousal and action units use the two-backbone specialist averages of the base system; expression averages the two shared-latent base members with the added parameter-isolated experts. For action units we additionally fit a per-unit decision threshold that maximizes each unit’s F1 on validation. This calibration is fit on validation and therefore optimistic on held-out data; a video-level cross-fit of the same threshold recovers only about a third of its apparent gain, so we report the calibrated and uncalibrated numbers separately and treat the uncalibrated 1.6949 as the robust number.

III. EXPERIMENTS

A. Evaluation protocol

The track scores a submission by summing the three tasks’ native metrics,

$$P_{\text{MTL}} = \frac{\text{CCC}_v + \text{CCC}_a}{2} + \frac{1}{8} \sum_{c=1}^8 \text{F1}_c^{\text{expr}} + \frac{1}{12} \sum_{k=1}^{12} \text{F1}_k^{\text{au}} \in [0, 3], \quad (4)$$

the mean valence–arousal concordance plus the macro-averaged expression and action-unit F1. We use the official s-Aff-Wild2 MTL partition [2]–[7], [25]–[36], which is video-disjoint between training and validation, and report on the validation split. All model selection (hyperparameters, early stopping, member admission, and thresholds) uses only the training and validation splits. Because the effective sample size at this granularity is the number of videos rather than frames, we size every reported difference with a paired bootstrap over the fifty validation videos and report the one-sided p-value; a gap that falls inside this noise is reported as positive-within-noise rather than as a significant gain.

Implementation details. The two ViT-B/16 backbones are AffectNet-initialized and adapted per-frame on s-Aff-Wild2 with the top 40% of blocks unfrozen, for three epochs at batch size 64 and learning rate 5×10^{-4} . Low-rank experts use rank 16, scaling factor 32, and dropout 0.1 on the attention and MLP projections. All temporal heads are bidirectional GRUs with hidden size 256, two layers, and dropout 0.2, applied over windows of 64 frames at stride 32 and trained for up to 25 epochs at learning rate 10^{-3} ; the shared affect-latent uses a 96-dimensional bottleneck with a KL weight of 0.05 annealed over the first eight epochs. Each member is a seed average, ten seeds for the primary shared-latent member and six for the experts and specialists.

B. Main result

Tab. I reports the validation cumulative results, on which the two-backbone shared-latent base system scores 1.6669. Adding the AffectNet parameter-isolated expert to the expression ensemble raises expression macro-F1 from 0.4698 to 0.4978 and the total to 1.6949, our robust primary result;

TABLE I

VALIDATION LADDER ON S-AFF-WILD2 MTL; EACH ROW ADDS ONE COMPONENT. THE SHARED-LATENT TWO-BACKBONE BASE FOLLOWS COMPANION WORK [1]; ROWS BELOW ARE CONTRIBUTED HERE. PER-AU CALIBRATION IS DEV-FIT. BEST P_{MTL} IN BOLD.

System	VA	EXPR	AU	P_{MTL}
Organizer baseline (ConvNeXt, MixAugment) [3]	–	–	–	0.45
Two backbones (FSFM + MAE-Face), SAL EXPR	0.6407	0.4698	0.5564	1.6669
+ AffectNet-LoRA expert (SAL EXPR)	0.6407	0.4978	0.5564	1.6949
+ per-AU threshold calibration	0.6407	0.4978	0.5699	1.7084
+ FSFM-LoRA expert	0.6407	0.4993	0.5699	1.7099
+ MAE-LoRA expert	0.6407	0.4970	0.5699	1.7076
+ pool SAL members' VA into VA ensemble	0.6567	0.4993	0.5699	1.7259

TABLE II

EXPRESSION MEMBER STRENGTH (MACRO-F1 ALONE) FOR ADMITTED MEMBERS.

EXPR member	role	macro-F1
FSFM SAL	base member	0.4461
MAE-Face SAL	base member	0.4416
AffectNet-LoRA	isolated expert	0.4762
FSFM-LoRA	isolated expert	0.4451
DINOv2 curriculum	curriculum near-peer	0.4540

the paired bootstrap places the added expert above the two-backbone expression ensemble by a positive-within-noise margin at 50 videos. Per-AU calibration adds a further dev-fit gain on action units, from 0.5564 to 0.5699, and a second low-rank expert brings the system to 1.7099, the point of diminishing returns for expression members. The diversity axis offers one further member: the shared-latent members also produce a valence–arousal prediction as a byproduct, and pooling it into the valence–arousal ensemble, alongside the two dedicated specialists, forms a further decorrelated near-peer. It raises the mean CCC from 0.6407 to 0.6567, the one member gain to clear significance under the paired video bootstrap, at a p-value of 0.005, and lifts the overall score to 1.7259, the strongest configuration on validation.

C. The strength and diversity axes

Tab. II scores each expression member on its own, and Tab. III reports the pairwise prediction correlation. Together they place the admitted members in the useful region of both axes: the two base members, the FSFM-LoRA expert, and the curriculum near-peer sit at macro-F1 0.4416–0.4540, the AffectNet-LoRA expert leads at 0.4762, and all of them are decorrelated enough to average well. The correlation table summarizes the mechanism behind parameter-isolation in a single line, plotted in Fig. 2: full fine-tuning curricula on one backbone sit at a prediction correlation of 0.98, low-rank experts on the same backbone at 0.91, and the strongest expert against a different backbone at 0.75, comparable to the 0.81 between the two base backbones themselves.

IV. CONCLUSION

Ensemble construction on this benchmark proceeds one member at a time, and the operative question is which member

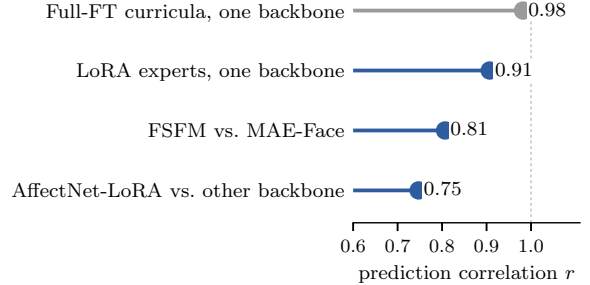


Fig. 2. Prediction correlation between expression members; lower correlation is more fusable. Full fine-tuning of one backbone re-converges at 0.98, whereas parameter-isolation and a second backbone stay decorrelated.

TABLE III

PREDICTION CORRELATION: FULL FINE-TUNING RE-CONVERGES, LOW-RANK ISOLATION DOES NOT.

Member pair (all SAL EXPR)	pred. correlation r
Full-FT curricula, one backbone	0.98
LoRA experts, one backbone	0.91
AffectNet-LoRA vs. a different backbone	0.75
FSFM vs. MAE-Face (two backbones)	0.81

to add. We answered it with a strength-parity rule, requiring an added member to be both decorrelated and a near-peer, and with the mechanism the rule needs, parameter-isolation, which manufactures decorrelated near-peers from a single backbone where full fine-tuning cannot. On the 11th ABAW MTL track the resulting system reaches 1.6949 on validation and 1.7259 in its strongest configuration. The rule also implies its own limit: once the pool holds a few decorrelated near-peers, any further member is almost always correlated or weaker and the score plateaus, so closing the remaining gap will depend on a genuinely different near-peer representation rather than on new ways of averaging the existing members.

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